

Message-capacity limits of individual agents set the transition point of collective truth-finding in language-model networks

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Abstract

Human or large language model (LLM), an agent in discussion processes only a few of the others' contributions, bounded by cognition or context and cost. LLM collectives can settle on a wrong consensus even from a correct majority; we ask how far that bound alone, as a crowding parameter α , fixes their fate: each agent accepts a further input with probability $e^{-\alpha r}$ after accepting r , generating the network. Over 31,824 randomized queries, we found that an 8-billion-parameter model's judgment of a claim effectively reduces to a logistic function of a weighted sum of its inbox, equivalent to a stochastic binary neuron with divisively normalized weights. From these coefficients and degree statistics alone, the wrong-consensus basin should vanish at $\alpha^* = 0.435$ (fan-in 6.4 of 31 sources). In 1,414 episodes with assigned opinions the prediction failed: collectives reached the wrong consensus from almost any start, even 75%-correct majorities. Replaying every inbox through the measured response traces the failure to coefficient transport: claim-resolved coefficients reproduce the phase, matching observed attractors within 0.005. Reversing the claims' wording identified the driver: wording tilts every reading toward a claim's assertion, not its truth. On a second 8B model the pipeline predicted claim-dependent bistability; basin boundaries appeared where computed (one claim mispredicted) and eight-claim calibration matched predictions in 14 of 16 conditions. At 70B the assertion bias vanishes. A collective's fate is thus fixed by two single-agent measurements: the tilt a claim's wording induces, and the message capacity that sets the transition point.

Significance statement

Language-model agents are beginning to operate in groups that debate and settle on answers, and such groups can talk themselves into the wrong one. This work shows that a group's fate is largely set by two quantities measurable on a single agent. An agent's judgment reduces to the input-output rule of a stochastic binary neuron with divisively normalized inputs, so the group's transition point follows in closed form from how many sources each member reads; the second quantity is the tilt a claim's wording induces on each reading. Across three models, collectives followed the assertion rather than the truth in one and crossed predicted transition points in another. AI collectives can be assessed, and designed, before they run.

1 Introduction

Groups of LLM agents that exchange messages and update binary opinions are now studied as statistical-mechanical systems. [El et al. \(2026a\)](#) showed that a kinetic Ising model fitted to one-step transitions describes more than 10,000 such communities ($N = 32$, eight rounds), predicts individual trajectories, and generalizes to unseen communication graphs. In that framework, as in most multi-agent opinion studies, the interaction graph is supplied by the

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experimenter; the question of where the graph comes from, and what its statistics do to the collective outcome, is listed by those authors as future work.

A separate line of work on synaptic resource allocation provides a one-parameter answer to the first half of that question. If a receiving unit accepts a further input with probability $e^{-\alpha r}$ once r inputs are already accepted, the number of accepted inputs grows only logarithmically in the candidate pool: the mean in-degree is $(\log N)/\alpha$ with bounded variance, the full in-degree distribution $P_{\alpha,N}(k)$ obeys a closed recursion, and the distribution is invariant under reordering of the candidates (Fukushima, 2026). The parameter α — the crowding parameter — compresses everything about the acceptance process that matters for degree statistics. Throughout, α describes the crowding of a finite intake budget — the message capacity of an agent — at the level of whole messages; it is unrelated to the attention mechanism inside transformer architectures, and we avoid the word “attention” for this reason.

The bound itself is not particular to machines. Human deliberation operates under cognitive limits of the same shape — working memory holds only a few items at once (Cowan, 2001), and divided attention caps how many contributions a participant actually processes (Hodas and Lerman, 2012) — so a theory that turns an intake bound into a collective outcome is, at least potentially, a statement about committees and juries as well as about machine collectives. In human groups, however, neither the bound nor the network it induces can be measured or controlled; in LLM collectives, both can.

These two ingredients suggest a quantitative question. Consider a population in which a fraction x_0 of agents initially holds the correct answer to a binary question. After several rounds of exchange the population typically reaches a correct or a wrong consensus; the probability of the former, $q(x_0)$, is the committor of the collective dynamics. Mean-field theory on the generated networks predicts that the basin of the wrong consensus disappears entirely above a critical crowding level α^* : for $\alpha > \alpha^*$, even small correct minorities propagate. Our question is to what extent α^* can be computed in advance — from the degree statistics that α induces and from response coefficients measured on single agents in isolation — with no quantity fitted to collective runs. The nearest precedents come from LLM naming games, where emergent conventions and collective bias are established phenomena (Flint Ashery et al., 2025; Flint et al., 2026): a mean-field consensus condition has been assembled from single-token acceptance rates measured on isolated models (De Nobili et al., 2026), and a mean-field account of population-size-dependent collective bias from logit-extracted single-agent policies (Flint et al., 2026). These constructions share our logic, but they concern coordination on a convention — no correct side, no committor — under fully mixed or imposed interaction structures. No previous study has, to our knowledge, generated the interaction network of an LLM collective from a crowding process, or predicted the basin boundary of a truth-finding collective — the critical initial fraction of correct agents — from single-agent measurements frozen before the collective ran.

Two families of confound make this comparison easy to get wrong, and both have shaped our design. First, in observational data the initial fraction x_0 is not an experimental dial: we show in Section 3 that in the largest public dataset x_0 is driven by question difficulty, which also drives the outcome, so the apparent committor is a difficulty curve and the transition point (the tipping point of the social-dynamics literature) is unidentified. Our collective experiment therefore assigns initial opinions exogenously. Second, changing the network changes the composition of each agent’s prompt, and LLMs respond to prompt composition regardless of any network; effects of this kind reproduce in a single agent with a single query. We therefore measure the single-agent response function first (Section 4), predict the collective outcome from it, and treat any residual — quantified by replaying every realized prompt through the measured response function — as the collective contribution proper.

The paper proceeds in three steps: a closed-form theory of α^* (Section 2), the measurements that reduce the experimental agent to a stochastic binary neuron and give the theory its

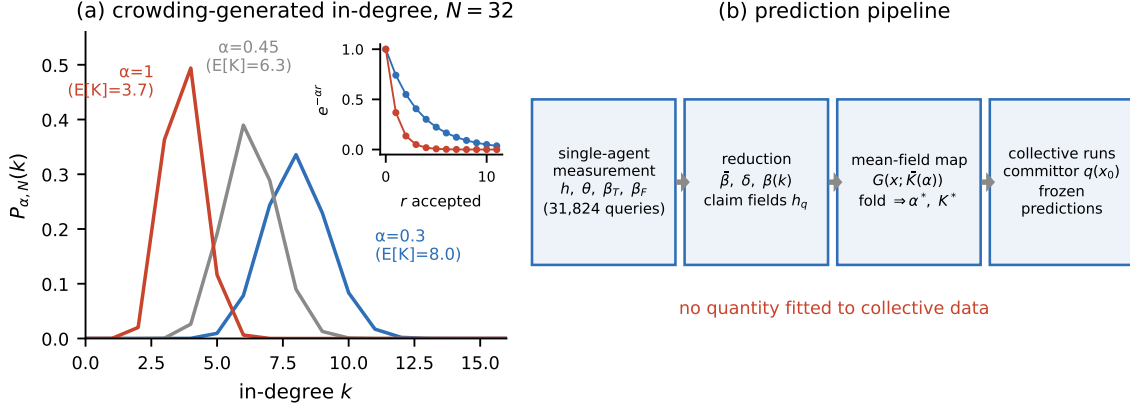


Figure 1: **Crowding-generated networks and the prediction pipeline.** (a) In-degree distribution $P_{\alpha,N}(k)$ generated by the acceptance process $e^{-\alpha r}$ (inset) at three crowding levels, $N = 32$. (b) Every quantity used to predict the collective outcome is measured on single agents or computed in closed form; nothing is fitted to collective data.

empirical coefficients (Sections 3–4), and a collective experiment run against the frozen prediction (Section 5). The pre-registered prediction fails, and the failure is traced to a measurable aggregation error rather than to the collective step. Two further pre-registered campaigns then identify what dominated: an assertion-polarity bias specific to the model class, which a second model does not share — and in that second model the same pipeline predicts claim-dependent bistability whose basin boundaries are observed at the blindly computed positions. To what extent a collective’s truth-finding is decided by individual input crowding is, on this evidence, a question with a computable answer — model by model, claim by claim.

2 Theory: a closed-form critical crowding level

2.1 Network generation

The generation process and its degree statistics are those of Fukushima (2026): each receiving agent processes the other $N - 1$ agents in arbitrary order and, having already accepted r sources, accepts the next with probability $e^{-\alpha r}$ (the first is always accepted); the in-degree distribution $P_{\alpha,N}(k)$ follows the recursion $P_{m+1}(r) = P_m(r)(1 - e^{-\alpha r}) + P_m(r-1)e^{-\alpha(r-1)}$, with mean $E[K] = (\log N)/\alpha + O(1)$. New here is only the closure: the deterministic continuum limit gives the finite- N form

$$\bar{K}(\alpha) = \frac{1}{\alpha} \ln(1 + \alpha(N - 1)), \quad (1)$$

which underestimates the exact mean by 3% ($\alpha = 0.3$) to 11% ($\alpha = 2$) at $N = 32$; where precision matters we invert the exact recursion instead. Degree statistics are invariant under candidate reordering, so manipulations of *who* is read change nothing in $P(k)$ — the property that lets order-based conditions be compared at exactly matched communication budgets.

2.2 Single-agent response and its reduction

Measurements in Section 4 show that an agent receiving l correct-side and $k - l$ wrong-side messages adopts the correct side with probability $\sigma(c_0 + \theta \mathbf{1}[\text{own state correct}] + \beta_T l + \beta_F(k - l))$, where σ is the logistic function (with the per-message weights attenuated by inbox size as

$\beta g(k)$; Section 4); the measured unit is equivalent to a stochastic binary neuron. Three combinations organize everything that follows: the symmetry-breaking field $h = c_0 + \theta/2$ (the persistence indicator decomposes into a constant field $\theta/2$ plus a symmetric term), the per-input swing $\bar{\beta} = (\beta_T - \beta_F)/2$, and the per-input drift $\delta = (\beta_T + \beta_F)/2$. The population dynamics under heterogeneous mean field (HMF) — each of an agent’s k inputs correct independently with probability x — is the one-dimensional map

$$F(x) = \sum_k P_{\alpha,N}(k) \sum_{l=0}^k \binom{k}{l} x^l (1-x)^{k-l} [x \sigma(u_1) + (1-x) \sigma(u_0)], \quad (2)$$

with $u_s = c_0 + \theta s + \beta_T l + \beta_F(k-l)$; the interior unstable fixed point of F separates the two consensus basins. The construction is the mean-field analysis of the generating model (Fukushima, 2026) with one substitution: the hard-threshold unit assumed there is replaced by the measured logistic unit, of which the threshold unit is the infinite-gain limit; the committor observable $q(x_0)$ is likewise inherited from that analysis, which computed it by an absorbing Markov-chain closure. No threshold nonlinearity is assumed anywhere below — bistability is a property of the S-shaped map, which crosses the diagonal three times whenever the linear gain defined in Section 2.3 exceeds one — and the substitution removes the structural discontinuity that hard-threshold updates produce at truth-weight 1 (equal per-message weight for correct-side and wrong-side sources; Section 3). Exact numerics iterate Eq. (2) directly; closing the degree at $\bar{K}(\alpha)$ and smoothing the binomial with the standard Gaussian–logistic approximation yields the closed-form reduced map

$$G(x; K) = x \sigma(z_+) + (1-x) \sigma(z_0), \quad z_s = \frac{(h \pm \theta/2) + K\delta + 2\bar{\beta}K(x - \frac{1}{2})}{\sqrt{1 + \frac{\pi}{8} 4\bar{\beta}^2 K x(1-x)}}. \quad (3)$$

A degree-dependent coupling $g(k)$ — for example the divisive normalization $\beta/(1 + \gamma k)$ that our data favor — enters by multiplying $\bar{\beta}$ and δ , leaving the form unchanged.

2.3 The critical capacity

α^* is the capacity at which the interior unstable fixed point of G undergoes a fold ($G = x$ and $G' = 1$ simultaneously). Operationally: the fold defines a threshold degree K^* , and α^* solves $\bar{K}(\alpha) = K^*$. The mechanism is visible in the linear gain at the symmetric point,

$$\lambda(K) = [\sigma(z_+) - \sigma(z_0)] + \frac{2\bar{\beta}K \bar{\sigma}'}{\sqrt{1 + \frac{\pi}{8} \bar{\beta}^2 K}}, \quad (4)$$

a persistence gain (first term, carried by θ) plus a social-amplification gain that grows only as $\sqrt{K}$ because input fluctuations smooth the response; bistability requires $\lambda > 1$. The $\lambda = 1$ criterion locates the mechanism but overestimates α^* by 30–90% at realistic fields, so all quantitative predictions below come from the fold of G . One further structural fact falls out of the reduction: with the measured $\delta < 0$ the effective field $h + K\delta$ is screened as degree grows (it crosses zero near $K \approx 10$ for the coefficients measured in Section 4 under the collective experiment’s prompt, variant A), which is why bistability survives at low α despite $h > 0$.

Against exact numerics (the full $P(k)$ sum, fold located by scanning α at 0.02 resolution) the closure is accurate to a median 2.1% over a 3-decade parameter grid and both measured coefficient sets when the exact $E[K]$ is inverted, and 6.4% in the fully closed form; the worst case (a saturating rule whose $K^* \approx 3$ strains the Gaussian smoothing) is 9% (Fig. 2). For the measured variant-A coefficients the prediction is $\alpha^* = 0.435$, $K^* = 6.4$: the wrong-consensus basin is predicted to disappear when agents accept, on average, fewer than 6.4 of their 31 possible sources. The critical crowding level of the collective is, to this accuracy, a closed-form function of single-agent coefficients.

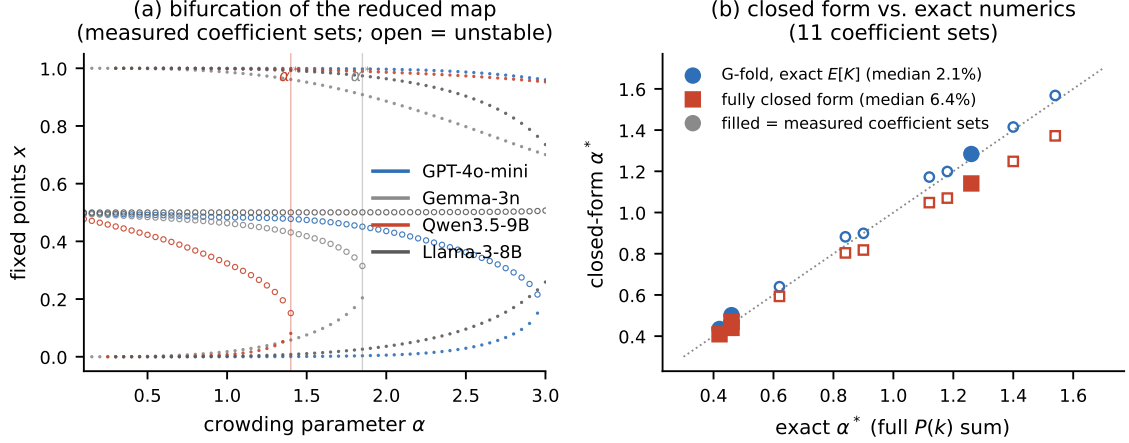


Figure 2: **Closed-form critical crowding level.** (a) Fixed points of the reduced map G as functions of α for the four measured coefficient sets of El et al. (2026a); the interior unstable branch (open symbols) vanishes at α^* . (b) Validation of the two closure levels against exact numerics over 11 coefficient sets (filled symbols: coefficient sets measured in this study).

3 Why public data cannot test the prediction

The natural first test requires no new data: El et al. (2026a) released full logs of 6,400 objective-question communities (four models, 40 questions, ten graphs per question, four episodes) (El et al., 2026b). We computed each community’s correct-side fraction trajectory $x(t)$ and asked whether the apparent committor $q(x_0)$ identifies a transition point.

It does not, for a reason that constrains all observational studies of this type. Across the four models, between 50% and 99.5% of the variance of x_0 lies *between questions*; within questions the x_0 -outcome correlation drops to 0.04–0.27. Communities at $x_0 \approx 0.40$ on different questions run to opposite consensus with near certainty, while communities at $x_0 = 0.15$ on an easy question converge correctly in 32 of 32 episodes. Question difficulty drives both the initial fraction and the outcome; the pooled $q(x_0)$ curve is a difficulty curve, and its 0.5-crossing is not a property of the collective dynamics. A question-clustered bootstrap widens the apparent confidence intervals on the fitted crossing by a factor of 4–10, after which no pairwise difference among models or graph families survives. Two further obstacles are structural: the released graphs have nearly degenerate degree distributions (random family mass concentrated on $k = 2$ –4; lattices), so the degree-statistics content of any mean-field prediction is untestable; and under the hard-threshold update rule assumed in earlier work the predicted crossing jumps discontinuously at truth-weight 1, reducing the comparison to a single bit. These are limitations of the identification strategy, not of the dataset’s quality — the same logs support the message-level measurement below, which is what makes the rest of this paper possible.

The logs do yield the microscopic coefficients. Regressing each agent’s next-round correctness on the composition of its inbox (43,500 observations per model, question-clustered errors), the truth asymmetry w_T — the relative per-message weight of correct-side sources, previously estimated at 1.16–1.71 from Ising couplings — is indistinguishable from 1 in all four models (e.g. GPT-4o-mini: 1.055, 95% CI [0.996, 1.114]), while three other coefficients are robustly nonzero: a persistence term θ , an individual field c_0 , and an attenuation $\lambda_{\text{lab}} = 0.08$ –0.56 of messages arriving through “disagree”-labeled edges. The asymmetry that breaks the symmetric degeneracy of the mean-field map therefore resides in the field $h = c_0 + \theta/2$, not in message weighting — which is what Section 2 assumes. Public data fix the form of the

theory; they cannot test its collective prediction.

4 Measuring the response function of the experimental agent

The collective experiment uses llama3.1:8b (4-bit-quantized (Q4_K_M), temperature 0.7, snap-judgement prompt matching El et al.’s protocol). We measured its response function directly: 31,824 single-shot queries in which the inbox composition (k messages, l of them correct-side), the A/B label assignment, the message order, and an optional explicit own-state line were randomized by design. Claims came from CLIMATE-FEVER (Diggelmann et al., 2020); a two-stage calibration selected 17 claims with per-claim field $|h_q| < 0.4$ *under the full experimental prompt scaffold* — a detail that matters, because fields calibrated under a bare prompt did not transfer to the scaffold (per-claim intercepts re-expanded to a range of 3.5 logits). The selected claims are 16/17 REFUTES (claims labeled false in CLIMATE-FEVER); consequently all results are stated in correct-side/wrong-side terms, and the truth-versus-polarity distinction is explicitly not identified by this claim set: a diagnostic regression shows TRUE-side messages are uniformly more persuasive (ratio 1.44), which accounts for the apparent $w_T = 0.74\text{--}0.89 < 1$.

Three measured facts carry the predictions (Fig. 3; per- k coefficients and interval estimates in SI Fig. S1). First, the coefficients: without an explicit own-state line (variant A, the protocol of the collective experiment) $h = 0.725$ [0.516, 0.915]; with it (variant B) $\theta = 1.909$ [1.723, 2.106] — indicating that the persistence measured in observational data is not an in-prompt self-input but a persona-level carryover, since adding one line raises θ from absent to ≈ 1.9 . Second, the load attenuation: adding messages attenuates each message’s influence, and model comparison (cross-validated by claim) favors inbox-size-dependent forms — load decay $e^{-0.14k}$ and divisive $\beta/(1 + 0.9k)$ are indistinguishable — over the position-dependent form $e^{-\alpha_{\text{pos}}r}$ closest to the theory’s acceptance kernel. The theory’s kernel is thus implemented by the exogenous router, not by the model’s internal weighting; we had pre-registered this demotion, and the internal decay enters the theory only through $\beta(k)$. The reduction itself is quantified twice: by claim-clustered cross-validated log-likelihood (-0.483 nats per observation for the two retained forms, against -0.509 for a claim-fields-only null and -0.693 for chance), and by the calibration of the fitted σ on every replayed collective transition (expected calibration error 0.05–0.10; Section 5). The same reduction is visible directly: once the per-message weights are divisively normalized, the observed correct-side rate in every design cell lies on the logistic curve of its fitted net input (RMS deviation 0.054 across 78 cells), whereas with constant weights the cells fan out by inbox size (0.118): large- k cells lie between the curve and 0.5 and small- k cells beyond it (Fig. 3). Third, the predictions: pushing the measured coefficients through Section 2 gives $\alpha^* = 0.40\text{--}0.46$ for variant A *under every reduction of the k -dependence* (constant, divisive, and the raw per- k coefficients), while for variant B the reductions disagree — $\alpha^* = 0.46$ (constant) versus 1.27–1.33 (saturating). The disagreement is an experimental resource: an arm run under variant B between $\alpha = 0.6$ and 1.3 discriminates the coupling rule regardless of outcome. The response function of a single agent, measured in isolation, fixes every number the collective experiment is asked to reproduce.

5 Collective experiment

5.1 Design (pre-registered before any collective run)

Networks are generated per episode by the acceptance process at capacity α ; initial opinions are assigned exogenously ($\lceil 32x_0 \rceil$ agents seeded correct-side); dynamics follow El et al.’s proto-

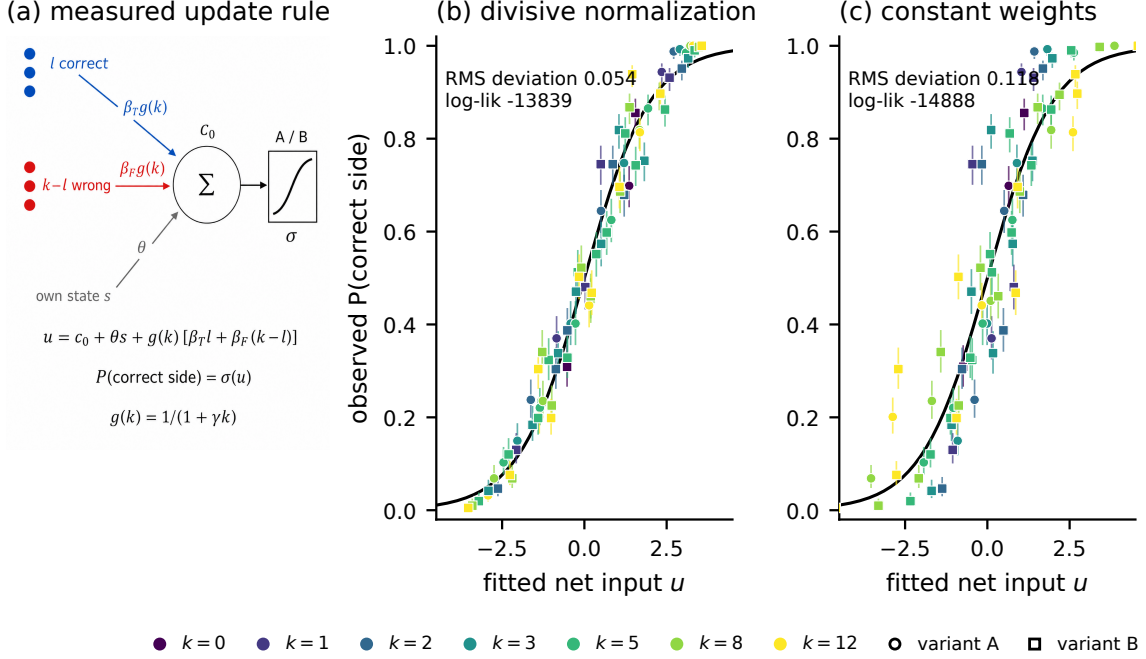


Figure 3: **Observed correct-side rate against fitted net input under divisive and constant message weights.** (a) The measured update rule: l correct-side and $k-l$ wrong-side messages enter with weights $\beta_T g(k)$ and $\beta_F g(k)$, the own-state indicator $s \in \{0, 1\}$ (variant B only) with weight θ , and the field c_0 ; the choice probability is $\sigma(u)$ with $g(k) = 1/(1 + \gamma k)$, $\gamma = 0.9$. (b) Observed correct-side rate in each of the 78 design cells of the 31,824 randomized queries (variant $\times k \times l \times s$; 408 queries per cell; Wilson 95% intervals) against the fitted net input u under divisive normalization, with claim fixed effects. Color gives inbox size k ; circles variant A, squares variant B. The curve is $\sigma(u)$, not a fit to the points. (c) The same cells under constant weights ($g \equiv 1$): large- k cells lie between the curve and 0.5 and small- k cells beyond it. RMS deviation of cell rates from $\sigma(u)$: 0.054 in (b), 0.118 in (c).

col — one broadcast message per agent per round conditioned on its current stance, Markovian inboxes, snap-judgement opinion resampling — for eight rounds. The main arm (variant A prompts) runs $\alpha \in \{0.20, 0.30, 0.38, 0.45, 0.55, 1.00\} \times x_0 \in \{4, \dots, 24\}/32$, adaptively concentrated where q crosses $\frac{1}{2}$ and, above the predicted α^* , at low x_0 where the collapse of the wrong basin is most visible. The discrimination arm (variant B prompts) runs $\alpha \in \{0.60, 0.90, 1.30\}$. A rewiring control (in-degree-preserving shuffle with rewiring fraction $\rho = 1$, i.e., all sources resampled, at $\alpha = 0.30$) tests the mean-field prediction that only $P(k)$ matters; a yoked one-step analysis replays every realized inbox through the Section 4 response function, so that any collective deviation from iterated single-agent behavior is measured rather than assumed absent; and a surrogate rollout of the measured per- k rule on the same graphs and seeds provides the finite-size prediction band. Decision criteria were fixed in advance: the empirical α_{emp}^* is the capacity at which the wrong-consensus rate from $x_0 \leq 8/32$ falls through 20%, and the discrimination arm’s verdict is read from which α levels retain a wrong basin.

5.2 Results

The pre-registered prediction failed, and we state the failure before its diagnosis. Across all 1,414 episodes (zero parse failures), the correct-consensus probability never reached 0.5 in any (α, x_0) cell: the collectives converge on the wrong consensus from almost every initial

condition, including $x_0 = 24/32$ (Fig. 4a). The wrong-consensus rate from low x_0 falls monotonically with α — 0.98 at $\alpha = 0.20$ to 0.75 at $\alpha = 1.00$, the direction the theory predicts — but never crosses the pre-registered 20% threshold, so no empirical critical crowding level exists in the window: the prediction $\alpha^* = 0.435 \pm 0.03$ is rejected (claim-clustered bootstrap, 0 of 2,000 resamples crossing).

The diagnosis localizes the failure. Replaying every realized inbox through the Section 4 response function calibrates well on average (Brier 0.10–0.16 against 0.17–0.25 for always predicting the base rate) but shows a -0.20 systematic wrong-ward error exactly at contested compositions ($l/k \approx \frac{1}{2}$); restricting the Section 4 *randomized* data to the four claims and compositions the collective experiment used reproduces the episode-level transition rates (correct-side probability 0.29 in the balanced cell, against 0.54 under the pooled coefficients), so the discrepancy exists already in single-shot data and is a failure of the pooled reduction — a claim-by-composition interaction that the shared coefficients with additive claim fields average away — not an emergent collective effect. We call this failure mode coefficient transport: coefficients measured under one claim mix do not carry over to another. Re-inserting claim-resolved coefficients into the unchanged theory predicts the observed phase: a single wrong-consensus attractor (stable fixed point x^*) at all α in the window, its position rising from $x^* = 0.011$ ($\alpha = 0.2$) to 0.078 ($\alpha = 1.0$), matching the observed monotone α trend of final states.

The discrimination arm survives this reanalysis and sharpens (Fig. 4b). With claim-resolved coefficients the two coupling reductions still separate at the discrimination-arm capacities: the constant rule predicts the attractor drifting to $x^* = 0.44$ by $\alpha = 1.3$ (mostly undecided outcomes), the normalized rule predicts $x^* = 0.067$ and 0.093 at $\alpha = 0.9$ and 1.3. Observed low- x_0 final states are 0.061–0.072 and 0.092–0.097 — agreement with the normalized rule to within 0.005, and a clear rejection of the constant rule. Degree-preserving rewiring ($\rho = 1$ at $\alpha = 0.30$) changes the committor by at most 0.09 in any cell and by 0.00 in four of six, consistent with the mean-field premise that only $P(k)$ matters. One residual remains open: at high x_0 ($\geq 20/32$) the collectives absorb wrong-ward faster than even the corrected surrogate predicts; candidate mechanisms (repeated exposure to the same senders across rounds, interactions beyond an additive claim field, finite-time relaxation) are recorded for follow-up. The collective experiment thus rejects the blind prediction, verifies each link of the reduction chain except coefficient transport across claims, and identifies the claim-induced field as the quantity that decided the outcome in every cell tested.

5.3 What the field is: a pre-registered discrimination experiment

The dominant field admits three interpretations, which we separated with predictions registered before data collection: an assertion-polarity bias (collectives drift toward affirming the claim, whatever its content), a truth asymmetry (wrong content itself is favored), and a parametric-prior pull (collectives drift toward the model’s stored world knowledge). Two designs separate them. First, collectives on three *true* claims (labeled SUPPORTS in CLIMATE-FEVER): polarity and prior predict correct-side convergence, truth asymmetry predicts wrong-side. Second, collectives on minimally negated rewrites of the four original misinformation claims — logical content preserved, assertion direction reversed, equivalence checked by model judgment and by hand: polarity predicts the drift reverses with the wording; both alternatives predict it does not.

The verdict is polarity (Fig. 5). On all three true claims the collectives converged on the correct (affirming) side from every initial condition, including 12.5% correct starts (mean final fractions 0.67–1.00), rejecting truth asymmetry. On all four negated rewrites the collectives converged on the affirming side (final fractions 0.98–1.00) where the identically contentful originals had converged on the denying side (0.00–0.43): the content-referenced drift reverses

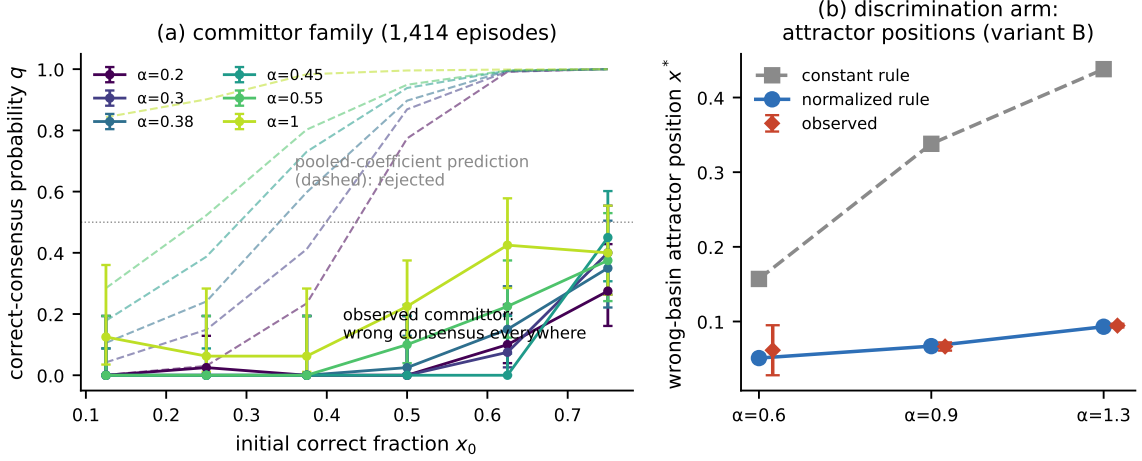


Figure 4: **Collective experiment against the frozen prediction.** (a) Observed committor family (solid, Wilson 95% intervals) against the pooled-coefficient surrogate (dashed): the blind prediction is rejected — wrong consensus from almost every initial condition. (b) Discrimination arm: observed wrong-basin attractor positions match the normalized (divisive) coupling rule to within 0.005 and reject the constant rule.

completely with the wording, rejecting the parametric-prior account. One rewrite whose bare-prompt field pointed to the *denying* side ($p = 0.34$) still converged on the affirming side, showing the polarity channel overriding the field rather than merely riding it. Blind single-agent predictions again matched the collective outcomes claim by claim, with one recorded exception (claim 1103 at $\alpha = 0.55$, the one cell where a negative field was predicted to beat the polarity channel and did not). For this model, the collectives follow the assertion, not the truth — which reframes the social reading of Section 5.2: misinformation loses here because misinformation is phrased as affirmation, not because wrong content is favored. The same reading bears on reports that LLM opinion networks converge toward scientific ground truth regardless of initial stance (Chuang et al., 2024): where the true position is the affirming one, a polarity channel and a truth bias make identical predictions, and only designs that reverse the assertion direction — as here — separate them.

5.4 The regime is a model property

Running the identical pipeline — claim calibration, randomized single-shot measurement, frozen fixed-point predictions, then collectives — on a second 8-billion-parameter model (qwen3:8b) produced a qualitatively different and sharper outcome (Fig. 6). For this model the theory predicted, before any collective run, claim-dependent bistability: an interior unstable fixed point at 0.12–0.15 for one claim and at 0.62–0.66 for another. Both appeared. Collectives seeded at $x_0 = 0.125$ split between the two basins while every start at 0.375 or above converged correctly, and — at the capacity where the predicted threshold is 0.62 — collectives seeded at exactly $x_0 = 0.625$ split nearly evenly, with 0.875 starts converging correctly. These are the first observed interior committor thresholds in this study, at the blindly computed positions. The predictions were wrong for one of the four claims (wrong-side dominance predicted, correct-side convergence observed), and partially wrong for a second at the lowest capacity; both misses are recorded alongside the hits. This model is also not assertion-following: one true claim converged to the *denying* side from all starts below 0.625, which no polarity account permits.

A pre-registered calibration campaign then took the comparison beyond these four claims.

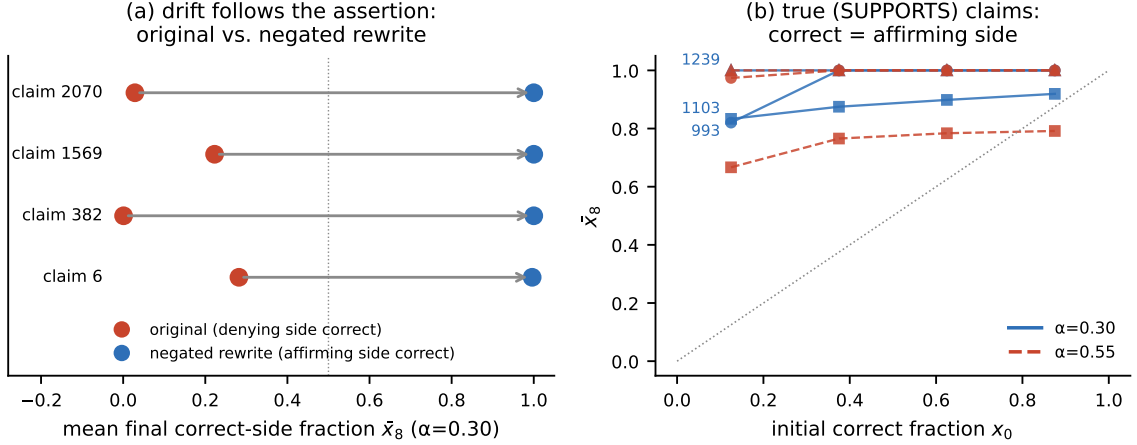


Figure 5: **Final correct-side fractions for original, negated, and true claims.** (a) Mean final correct-side fraction for the four original claims (denying side correct) and their minimally negated rewrites (affirming side correct) under identical conditions: the content-referenced outcome reverses with the wording. (b) True (SUPPORTS) claims converge on the correct = affirming side from all initial fractions.

Twenty further claims were calibrated and measured single-shot; the unchanged pipeline classified each (claim, α) cell — monostable correct-side, monostable wrong-side, or bistable with a computed threshold — and eight claims spanning the predicted regimes were selected by a rule fixed in advance. The full prediction table was written to file about one minute before the collective driver launched (SHA-256 recorded in the repository), and 256 further episodes were run ($\alpha \in \{0.30, 0.55\}$, eight initial fractions, zero parse failures). The predicted classification matched the observed one in 14 of 16 cells (Fig. 7); both mismatches fall on the single claim whose predicted threshold, 0.13, lies at the edge of the seeding grid, where the classification shifted by one category in each direction. Across the ten cells observed bistable, predicted and observed threshold positions are rank-correlated at $r_s = 0.68$ (Spearman, $p = 0.03$), with seven predictions inside the episode-bootstrap intervals; the one systematic deviation is a single claim whose collectives absorb correct-ward from initial fractions well below the predicted threshold (observed crossings 0.19 and 0.31 against predictions 0.46 and 0.54; full tables in the SI). What was an existence proof on two claims is thus a calibration curve on eight: the single-agent pipeline predicts not only whether a threshold exists but, to the resolution of the seeding grid, where it sits — and its errors are concentrated, identifiable, and one-sided toward the correct answer.

Scale closes the picture. In llama3.3:70b, probed at the single-agent level with the same claims and message bank, the per-message assertion asymmetry shrinks from 1.44 to 1.03 and the bare fields align fully with content ($|\text{logits}|$ 3.7–7.2, leaving no calibratable near-neutral claims): the polarity channel that governed the 8-billion-parameter collectives is absent, and its collective regime — not measured here, a stated limitation — would be field-dominated on any realistic claim. Whether a collective has a transition point at all, where it sits, and what force drives consensus are, across these three models, properties measurable on the individual agent before any collective is run.

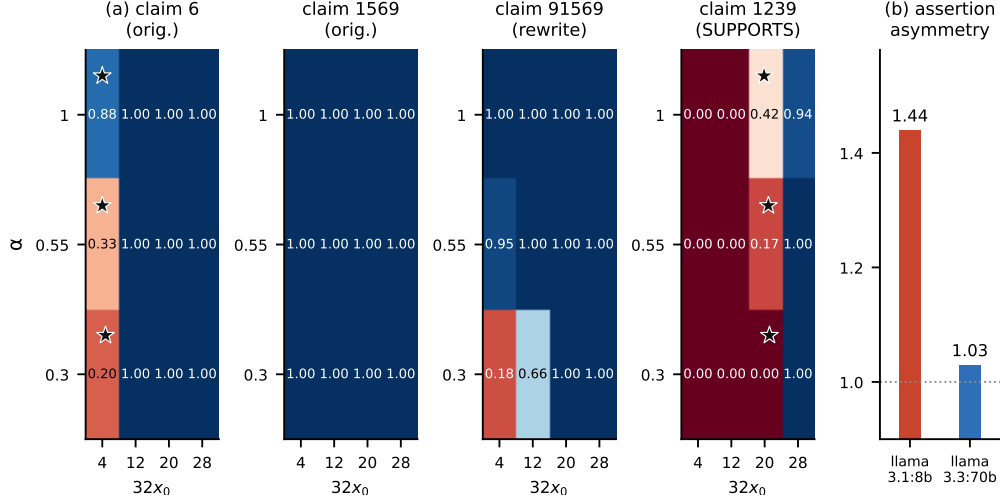


Figure 6: **qwen3:8b collective outcomes against predicted fixed points, and the per-message asymmetry at 70B.** (a) qwen3:8b collectives: mean final correct-side fraction per cell; stars mark the frozen predicted unstable fixed points, which coincide with the observed basin boundaries for claims 6 and 1239 (misses for claim 1569, and for 91569 at $\alpha = 0.30$, are reported in the text and SI). (b) The per-message assertion asymmetry vanishes at 70B scale.

6 Discussion

The intake budget is a design variable for collective reliability: through $P(k)$ it sets whether a wrong-consensus basin exists, and the critical budget K^* is computable in closed form from single-agent coefficients — on the precondition that the claim-induced field and any model-specific polarity channel are measured first, since these decide which phase diagram the budget is operating in. The evidence is three models run against frozen predictions: assertion-dominated monostability in llama3.1:8b, where field and polarity override the budget and 75%-correct majorities are lost; claim-dependent bistability with blindly predicted thresholds in qwen3:8b (14 of 16 cells); vanishing polarity at 70B. A blind prediction, its pre-registered rejection, the verified localization of the failure, and the cross-model confirmation of the corrected pipeline are reported together. Throughout, the α trend of the wrong-consensus attractor — observed and predicted — shows that the number of messages each agent was actually shown remained the operative degree even where the blind prediction failed: the failure mode was coefficient transport, not the identity of the degree.

What α corresponds to outside this experiment differs by system. Here α is an enforced reading budget: the router, not the model, implements the acceptance kernel, a division of labor the measurements themselves selected (Section 4). In deployed multi-agent systems its counterparts are the quantities that bound how many messages an agent reads — finite context windows, per-token costs, and the sampling policies of agent frameworks; in the naming-game literature the analogous bound is a finite interaction memory (Flint et al., 2026); in human information sharing, divided attention limits message visibility in the same direction (Hodas and Lerman, 2012). The model’s own capacity enters the theory separately, as the measured load attenuation $\beta(k)$: the intake budget sets who is read, the attenuation sets how much each reading counts.

Dialogue does not correct error in the field-dominated phase: majorities of 75% correct agents converged on the wrong consensus at every capacity tested, and the discrimination experiment shows the drive is the assertion, not the content. Communication volume and capacity are no remedy where the per-reading tilt is adverse — a limiting con-

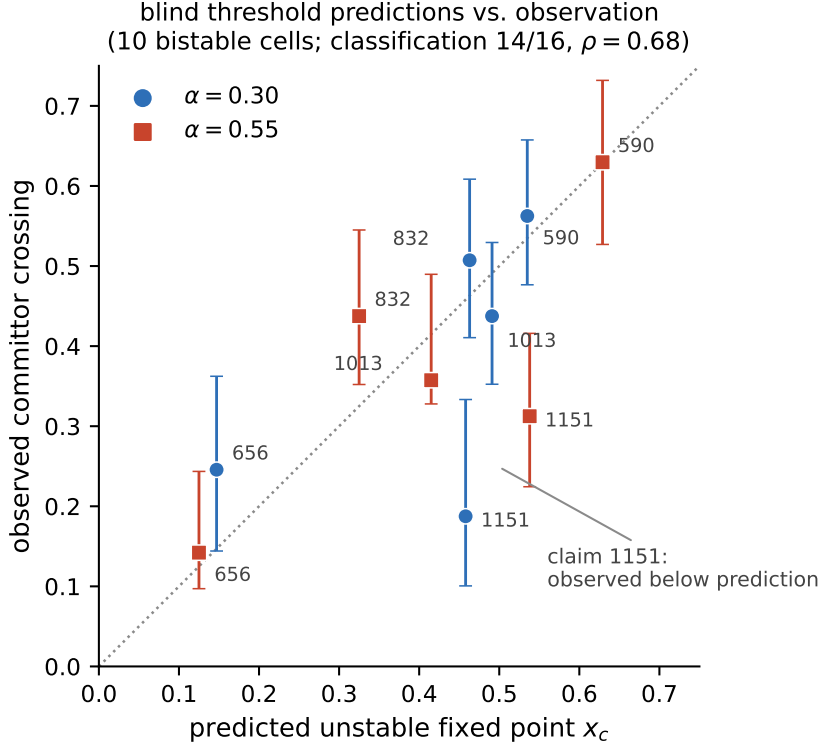


Figure 7: **Pre-registered eight-claim calibration of threshold predictions (qwen3:8b).** Predicted unstable-fixed-point position against the observed committor crossing for the ten (claim, α) cells observed bistable; vertical bars are episode-bootstrap 95% intervals, the dotted line is identity. The two cells of claim 1151 are the one systematic miss. Cells predicted and observed monostable (4 of 16) and the two boundary-grid mismatches of claim 1146 are listed in the SI.

dition on idealized-Bayesian accounts of collective knowledge (Taniguchi et al., 2025). The misinformation-vulnerability reading must be stated carefully: small llama collectives lose to misinformation because misinformation affirms; models without the polarity channel (qwen, 70B) fail or succeed by different routes. Directional conformity has likewise been reported to collapse majority-vote aggregation in LLM safety panels under peer pressure (Hu and Qu, 2026); the polarity channel measured here is a claim-level analogue that requires no peer opinions at all — it is present in bare single-shot calibration.

The origin of the polarity channel is probed and bounded rather than resolved. An ablated (refusal-direction-ablated, Arditi et al., 2024) derivative of the same instruct model retains the channel — per-message asymmetry 1.24 [1.05, 1.47] against 1.44 [1.25, 1.65] for the intact model, measured in the same batch with the same message bank — so removing refusal behavior does not remove the assertion bias, and any reduction is below this measurement’s resolution. Whether the channel exists before instruction tuning could not be measured: the base model does not follow the forced-choice format (parse rate below 5% even with a three-shot format prefix), which places pre-tuning behavior outside any forced-choice probe; a log-probability readout is the instrument for that question. The channel is adjacent to, but not identical with, sycophancy as usually defined — agreement with a user’s stated opinion, traced in part to the preference data used for RLHF (Sharma et al., 2023); here the agreement is with the claim’s own assertion, with no user opinion in the prompt. What can

be said is negative and useful: the bias that decides these collectives’ fate is not carried by the safety-refusal direction.

Two further single-agent observations are recorded without mechanistic claims: opinion flips concentrate in particular agents rather than being suppressed after a flip (a positive “anti-refractory” coefficient, $+0.98$), and the measured branching ratios (1.8–3.6) place these synchronous dynamics in a drift-dominated, far-from-critical regime — closing, for now, a self-organized-criticality reading of these collectives.

What is verified here is the reduction chain from single-agent response and degree statistics to a collective critical point in *LLM* collectives. The suggestion for human groups is nonetheless concrete. The cognitive limits cited in the Introduction give human participants a finite effective intake of the same shape, and the theory’s inputs are all behavioral — response tilt, load attenuation, intake bound — none tied to a machine substrate. If deliberating groups sit at finite effective α , the theory says which of them can recover the truth from a correct minority and which cannot. Statements about human collectives are implications mediated by one number: an estimate of a human α from intake-saturation data would locate human groups on the computed α^* curve, but no human committor has been measured here, and none ethically could be. A further limitation is the REFUTES-skew of the calibrated claim set, which restricts all claims to correct-side/wrong-side language; identifying genuine truth asymmetry needs a claim set balanced in polarity at matched difficulty, which this model’s one-sided accuracy on SUPPORTS claims may make impossible — with the mitigation that the α^* prediction never invoked truth asymmetry ($w_T = 1$ throughout).

Relative to [El et al. \(2026a\)](#), their graphs were given, ours are generated; their fits describe, ours predict before the run. The two studies probe orthogonal axes of one phase diagram for the same class of system — binary units with logistic (Glauber-type) updates. Their transition is thermal: topology fixed, noise swept, a critical point located by a susceptibility peak, on symmetric couplings that admit an energy function. Ours is topological: effective noise fixed (absorbed in the measured β), degree statistics swept through α , and the transition is a fold — the disappearance of a metastable basin, a spinodal rather than a critical point, which is why the committor collapse and not a fluctuation peak is the correct observable. Our graphs are directed, so the dynamics are non-equilibrium and free-energy arguments do not transfer; the mean-field map does. Model-dependent collective regimes have now been reported in the coordination setting as well — naming-game populations whose collective bias amplifies, induces, or reverses individual bias with model-specific size thresholds ([Flint et al., 2026](#)) — convergent with the model dependence measured here; and the structured interaction topologies those authors list as future work are what the crowding process supplies. For consensus-scaling observations ([De Marzo et al., 2024](#)), $(\log N)/\alpha$ supplies the analytic form for the reported size-limited consensus.

Part of the logistic form is guaranteed by the output mechanism: a forced choice between two options, sampled at temperature T from a softmax, makes the choice probability exactly a logistic function of the difference of two logits. What the mechanism does not guarantee — the empirical content of the reduction — is that this logit difference is approximately additive in the inbox composition, one weight per correct-side and per wrong-side message (Fig. 3b). Additivity in log-odds is what an aggregator of conditionally independent evidence would produce; whether the transformer approximates such a computation, and why, is left open alongside the mechanism of the attenuation below.

The measured update rule (Fig. 3a) departs from a classical Ising spin in exactly one place: the per-message coupling is attenuated by the total inbox size, $\beta/(1 + \gamma k)$, the form known in sensory neuroscience as divisive normalization ([Carandini and Heeger, 2012](#)). As in that literature, the law is phenomenological — a behavioral input–output relation, not an identified mechanism. Three candidate mechanisms make separable predictions: softmax attention inside the transformer is itself a divisive operation, and predicts attenuation governed by

token volume rather than message count; proportion coding — judging by the fraction l/k of supporting messages rather than their number, the saturation pattern classically reported for human majority influence — predicts dependence on l/k alone; and calibration compression of the output probabilities would mimic attenuation and is excluded or quantified by reading log-probabilities directly. A factorial manipulation of message length and proportional (l, k) scaling within the Section 4 protocol would separate the three; we leave this as a stated open question, since no result in this paper depends on which mechanism generates the measured form.

The practical rule this work supports is short: measure the tilt first, then set the budget. Both are single-agent measurements, and together they decide the collective’s fate.

Methods

Models and serving

All agents are served with ollama. The primary model is llama3.1:8b (Q4_K_M quantization, temperature 0.7 throughout, seed recorded for every call). The cross-model campaign uses qwen3:8b (tag 500a1f067a9f, thinking disabled via `think:false`) and llama3.3:70b (Q4_K_M; judgments only, message banks fixed to the 8B-generated banks). Single-shot campaigns ran against a user-space ollama instance with `OLLAMA_NUM_PARALLEL=16` (the system instance saturates at 0.62 generations/s; the parallel instance reaches 3.5 generations/s); the 70B model does not load under 16-slot KV allocation and was served by the serial system instance. The polarity-origin probe (Discussion) used two further variants, judgments only: llama3.1:8b-text-q4_K_M (the base model) and an ablated derivative of the instruct model (mannix/llama3.1-8b-ablated, Q4_0), with a three-shot answer-format prefix applied identically to all variants, including the re-measured instruct control. Every call’s prompt hash, seed, raw response, and parse result are logged to JSONL.

Prompts

The judgment prompt (verbatim; variant A) is:

```
# Your Expertise
{persona}
# Messages
You received the following messages from fellow panelists:
- {message 1} ...
# Problem
You are one member of an expert panel judging the following claim.
Claim: "{claim}"
Options:
A) {TRUE or FALSE}
B) {FALSE or TRUE}
# Instruction
Give your direct answer on which option is correct.
Do NOT work through the problem, show any steps, compute, or
explain --- just commit to a snap judgement.
## Response Format
Respond with a single character, A or B, and nothing else.
```

Variant B inserts, before `# Problem`: “`# Your Current Answer / Your current answer is that the claim is {TRUE|FALSE}.`” The A/B $\leftrightarrow$ TRUE/FALSE assignment is randomized per call and counterbalanced within cells; message presentation order is randomized and recorded. The message-generation prompt asks the sender, given its persona, the claim, and its current stance, to “write a brief message to a fellow panelist stating that you believe

the claim is {stance} and giving the single strongest reason from your reasoning,” with a two-sentence response format; stance is expressed in the message body (TRUE/FALSE), never as an option letter, because option labels are randomized per receiver. Personas are one-line expertise statements drawn from a fixed pool of 12. Prompt texts are extracted verbatim from the released scripts.

Claims and calibration

Claims are drawn from CLIMATE-FEVER (Diggelmann et al., 2020) (SUPPORTS and REFUTES only; the claim text is presented alone, without evidence). A two-stage calibration (24 calls per candidate, refined to 64 for candidates with $p_q \in [0.35, 0.65]$; A/B assignment counterbalanced) measured per-claim accuracy p_q and field $h_q = \text{logit}(p_q)$. The collective experiments use per-claim fields re-estimated under the full experimental scaffold (Section 4). The negated rewrites of the four Stage-2 claims (minimal negation; one De Morgan transformation to avoid a double negative) were checked for semantic opposition by model judgment (70–85% of 20 counterbalanced calls per pair) and approved by hand before any collective run. Full claim texts, rewrites, and calibration values are in the SI.

Networks, episodes, and controls

Each episode generates a fresh directed network: every receiver processes the other 31 agents in an independent uniformly random order, accepting the next candidate with probability $e^{-\alpha r}$ after r acceptances. Initial stances are assigned exogenously to a uniformly random subset of $\lceil 32x_0 \rceil$ agents. Each of eight rounds consists of a message phase (every agent with at least one out-edge generates one two-sentence message conditioned on its current stance, broadcast to all its receivers) and a judgment phase (each agent answers the snap-judgment prompt with its current-round inbox; single sample). Because the network is regenerated per episode and every message is regenerated each round from its sender’s current stance, correlations between a receiver’s inputs do not accumulate across rounds — the design property on which the mean-field independence assumption of Section 2 relies, and whose sufficiency the rewiring control tests empirically. A fidelity check re-ran ten episodes with five-sample majority voting; absorption was, if anything, sharper, so the single-sample simplification is not the source of wrong-side absorption. The rewiring control resamples each receiver’s sources uniformly while preserving its in-degree. Outcomes are classified at $\delta = 0.1$ (correct consensus: $x_8 \geq 0.6$; wrong: $x_8 \leq 0.4$), with $\delta \in \{0.05, 0.2\}$ and undecided-exclusion sensitivity analyses in the SI.

Cross-model calibration campaign

Twenty further CLIMATE-FEVER claims (ten SUPPORTS, ten REFUTES, disjoint from all earlier sets) were calibrated under the full scaffold and measured single-shot with qwen3:8b (24 calls per candidate, 64 in the boundary band; per-claim response-function fits from 6,720 further queries). The resulting fixed-point classifications and thresholds for $\alpha \in \{0.30, 0.55\}$ were frozen to file, and eight claims were selected by a rule fixed in the pre-registered instructions (at least four predicted bistable with interior thresholds, the remainder monostable, filled from the bistable pool where a category was empty — no wrong-side-dominant claim existed among the twenty candidates). Collectives ran the full grid $\alpha \in \{0.30, 0.55\} \times x_0 \in \{2, 6, \dots, 30\}/32$ with two replicates (256 episodes, protocol identical to the main cross-model campaign). Observed thresholds are the 50% crossings of episode-level outcomes against x_0 under a monotone logistic fit, with episode-bootstrap percentile intervals.

Statistics

All confidence intervals on measured coefficients are claim-clustered pairs bootstrap percentiles (1,000–2,000 resamples; claims are the clusters because episodes within a claim share its field). Attenuation-form comparison uses five-fold cross-validation with folds split by claim, alongside AIC. Per- k coefficients are unpenalized logistic fits within each k with claim fixed effects. For Fig. 3, one logistic fit per prompt variant with claim fixed effects and message weights $\beta_T g(k)$, $\beta_F g(k)$ (divisive, $\gamma = 0.9$ fixed, or $g \equiv 1$) assigns each query a predicted probability; a design cell’s net input u is the logit of its mean predicted probability, and its observed correct-side rate is plotted against u with a Wilson 95% interval. Fixed points and folds of the mean-field map are located by scanning x on a 1,200-point grid with bisection refinement, and α^* by scanning α at 0.02–0.05 resolution with bisection. Reanalysis statistics on the public dataset use question-clustered bootstrap and CR0 sandwich errors.

Pre-registration timeline

Throughout this paper, “pre-registered” means that a prediction, together with the criteria for judging it, was written to a file before the corresponding data were collected, and that the file’s timestamp and SHA-256 hash are recorded in the project repository, so that the order of prediction and data can be verified by anyone. No external registry (such as OSF) was used: the guarantee rests on the recorded timestamps and hashes, not on a third-party record. Four prediction sets were frozen, in writing, before the corresponding data existed: (i) the $\alpha^* = 0.435 \pm 0.03$ prediction and the decision criteria of the main collective experiment, before any collective episode (2026-08-23); (ii) the three-hypothesis prediction table of the polarity discrimination experiment, before its data collection (2026-08-25); (iii) the qwen3:8b fixed-point predictions, written to file 13 hours before the corresponding collective campaign completed (file timestamps verified; 2026-08-26); (iv) the classification and threshold predictions of the eight-claim calibration campaign, frozen to file (SHA-256 recorded) about one minute before the collective driver launched and six minutes before the first episode completed (2026-08-27). The one adaptive element (second-pass allocation of episodes) followed rules fixed in the pre-registered design.

Data and code availability

All episode logs (stance histories, full message texts, graphs, seeds, prompt hashes), single-shot logs, analysis scripts, and notebooks are retained in the project repository, to be released publicly on GitHub (https://github.com/dockmfgit/AI_agent_crowding); the public dataset of El et al. (2026a) is not redistributed and is available from its original source (El et al., 2026b). Layout: `scripts/` (measurement, engines, analysis), `results/` (all tables backing the figures), `data/` (raw JSONL and episode JSON), `notes/` (working records including the frozen prediction files).

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